Reproducing Table I of “Tracking down Quirks at the LHC”

Knapen, Lou, Papucci & Setford, arXiv:1708.02243 · short report · 3 October 2026

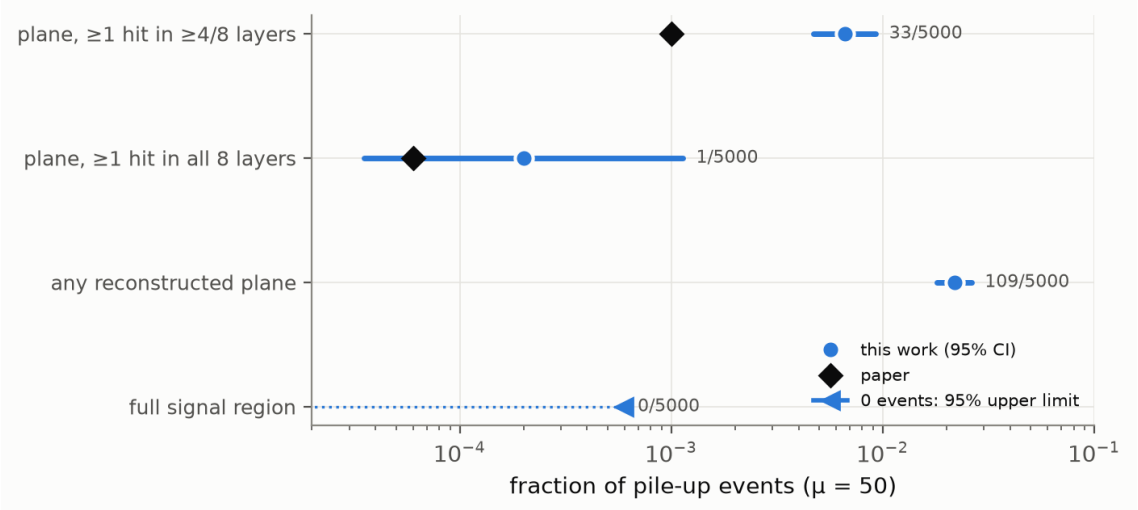
We redid the paper's signal and background study using only what the paper and its detector reference [27] actually say. Nothing was tuned to match the paper. Each simulated sample is shown on its own.

Exact p-value	What we call it
$p < 0.01$	**Strongly inconsistent**
$0.01 \le p < 0.05$	**Statistically inconsistent**
$0.05 \le p < 0.10$	**Borderline / mild tension**
$p \ge 0.10$	**Consistent**

How to read the tables. k/n = events passing / events tested. 95% CI = Wilson interval. With 0 events we quote the one-sided 95% upper limit (UL). p = exact binomial probability of our count if the paper's number were exactly right. The paper gives no event counts or errors, so its own simulation uncertainty is ignored; borderline p-values are therefore a little optimistic. “Paper” = what the paper says; “Our reading” = what we conclude.

1. Pile-up background (⟨μ⟩ = 50, 5000 events)

Quantity	Ours	Rate [95% CI]	Paper	p	Category	Our reading
Plane with hits in ≥ 4 of 8 layers	33/5000	6.6 [4.7, 9.3] $\times 10^{-3}$	$\sim 1 \times 10^{-3}$	9.8×10^{-17}	**Strongly inconsistent**	About 6× more planes than the paper
Plane with hits in all 8 layers	1/5000	2.0 [0.35, 11] $\times 10^{-4}$	6×10^{-5}	0.26	**Consistent**	Matches
Any reconstructed plane	109/5000	2.2% [1.8, 2.6]	not given	—	**n/a**	—
Full signal region	0/5000	UL 6.0×10^{-4}	“a few” in Fig. 5, judged removable	—	**n/a**	No fake signal, same conclusion



Pile-up rates with 95% intervals; diamonds are the paper's numbers.

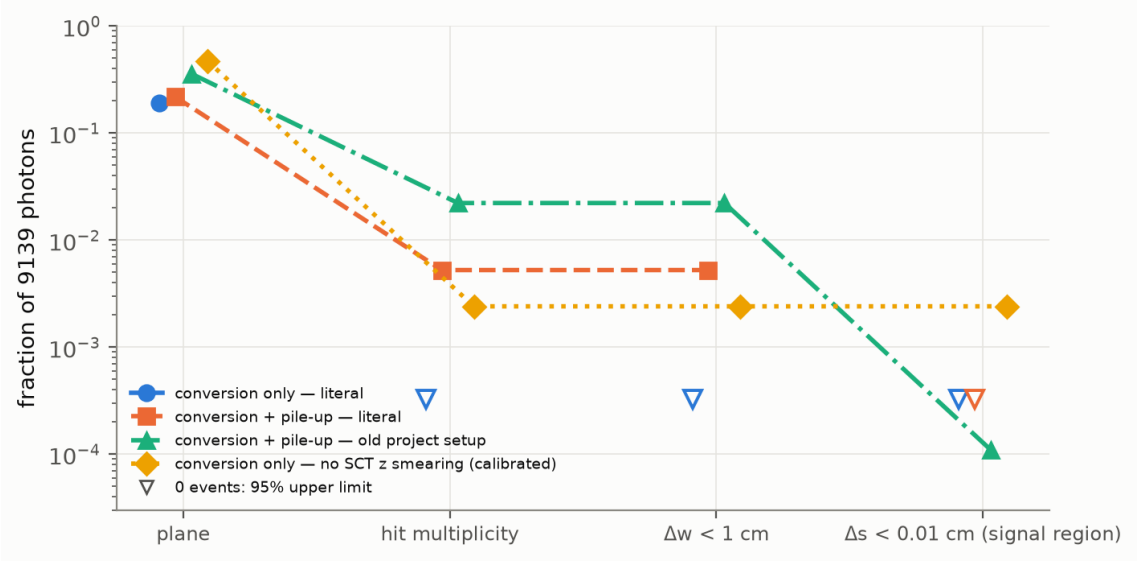
Why the 4-of-8 rate is higher

- Our pile-up tracks are ideal helices through 8 barrel layers. The paper used a fuller simulation (energy loss, bremsstrahlung, endcaps) that removes many soft tracks.
- Our minimum-bias pool has 20 000 events; the paper's had 125 000.
- The paper does not say whether “a plane” here means a seed or a fitted plane.
- The conclusion does not change: no pile-up event reaches the signal region.

2. Converted-photon background (9139 Z(νν)γ+jet photons)

Stage	Conversion only	Conversion + pile-up	Paper	Our reading
Reconstructed plane	1746/9139 = 19.1% [18.3, 19.9]	1990/9139 = 21.8% [20.9, 22.6]	not given	—
Hit multiplicity (7×2 + 1)	0/9139 (UL 3.3×10⁻⁴)	48/9139 = 0.53% [0.40, 0.70]	not given	Pile-up fills the gaps
Full signal region	0/9139 (UL 3.3×10⁻⁴)	0/9139 (UL 3.3×10⁻⁴)	negligible	**Matches**
Events at 300 fb⁻¹	< 0.21 (our σ) · < 0.98 (paper σ)	< 0.21 (our σ) · < 0.98 (paper σ)	≈ 0	**Matches**
Cross-section before conversion	0.23 pb		~1 pb	**Differs (×4.3)**

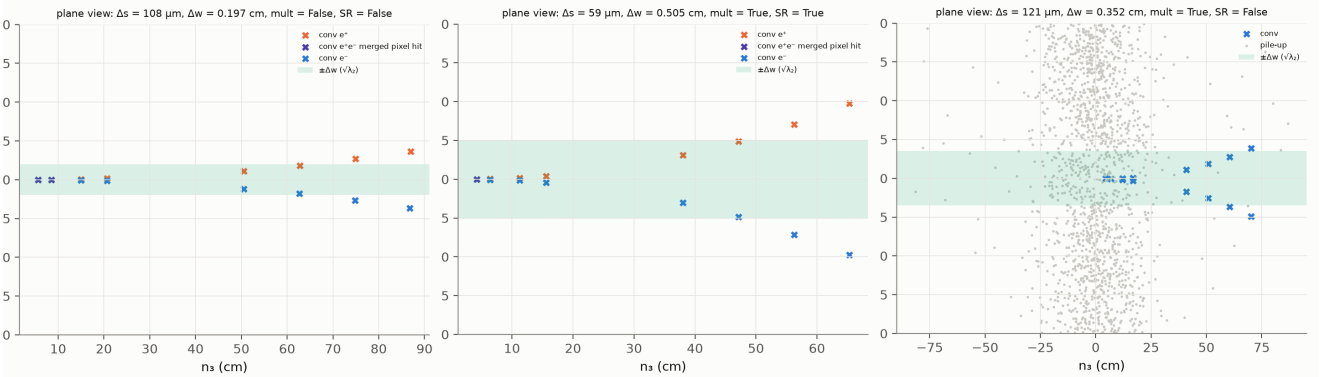
No p-value is possible for the background rows: the paper gives no rate or uncertainty to test against.



Fraction of photons surviving each cut. Open triangles: 95% upper limit when nothing survives.

What this shows

- With the paper's detector settings no conversion event fakes a quirk, with or without pile-up. The closest of the 48 overlay candidates has Δs = 121 μm against the 100 μm cut.
- The single event that passed in our old project setup (event 2555) does not even form a plane in the literal rerun.
- If the SCT smearing along the beam is switched off (the setting that rescues the signal efficiency, section 5), 22/9139 = 0.24% [0.16, 0.36] conversions pass: about 1.5–7 events at 300 fb⁻¹. That detector choice is not background-free.
- Our MadGraph cross-section is 4.3× below the paper's ~1 pb. The paper's generator settings are not given, so we show yields for both and never force 1 pb.
- Conversion probability 1% comes from the paper's own "~1 pb → ~10 fb"; the beam-pipe material needed for the PDG curves is not stated.



Hits in the plane's own coordinates (band = ±Δw). Left: conversion, literal (fails hit count). Middle: conversion without SCT z smearing (passes everything). Right: conversion + pile-up, closest literal candidate (Δs = 121 μm, just fails). Full displays: FINAL_FIGURES/F10.

3. Colored quirk signal (m_Q = 1800 GeV, Λ = 2 keV)

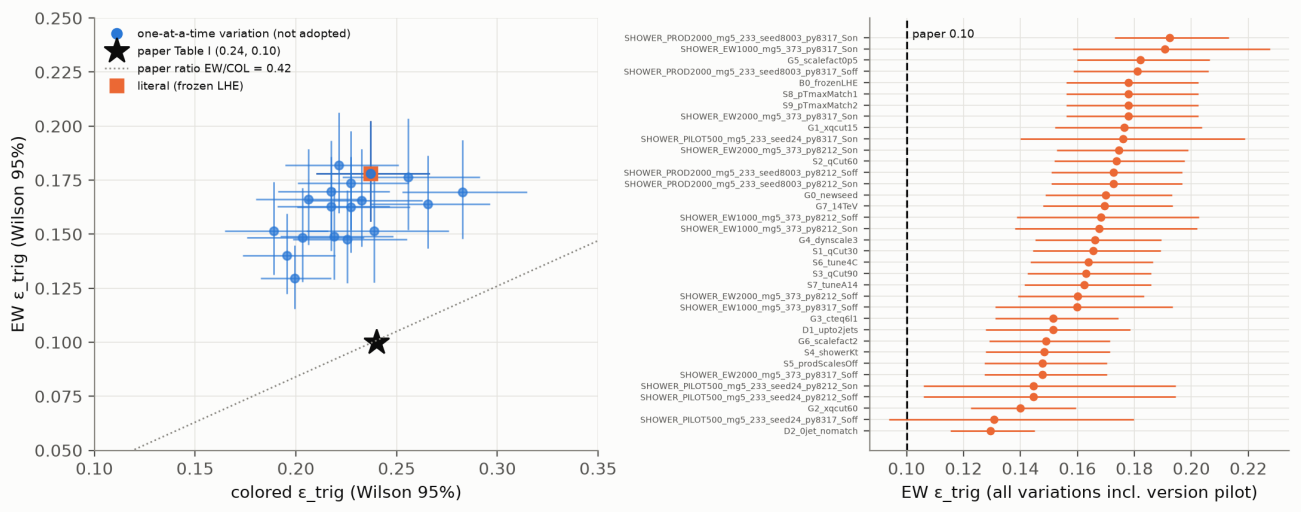
Efficiency	Sample	Ours	%	95% CI (%)	Paper	p	Category	Our reading
Trigger	COL1000	101/419	24.1%	[20.3, 28.4]	0.24	0.95	**Consistent**	Matches
Trigger	COL2000	207/873	23.7%	[21.0, 26.6]	0.24	0.87	**Consistent**	Matches
Fiducial	COL1000	6/27	22.2%	[10.6, 40.8]	0.28	0.67	**Consistent**	Matches
Fiducial	COL2000	22/68	32.4%	[22.4, 44.2]	0.28	0.42	**Consistent**	Matches
Reconstruction	COL1000	1/6	16.7%	[3.0, 56.4]	0.35	0.67	**Consistent**	Only 6 events: cannot tell
Reconstruction	COL2000	1/22	4.5%	[0.8, 21.8]	0.35	0.0013	**Strongly inconsistent**	Clear miss
Reco, no SCT z smearing*	COL1000	5/6	83.3%	[43.6, 97.0]	0.35	0.02	**Statistically inconsistent**	Diagnostic only
Reco, no SCT z smearing*	COL2000	8/22	36.4%	[19.7, 57.0]	0.35	1.00	**Consistent**	Diagnostic only

* Not in the paper. Shown only to measure how much one undocumented setting matters (section 5).

4. Electroweak quirk signal (m_Q = 800 GeV, Λ = 4 keV)

Efficiency	Sample	Ours	%	95% CI (%)	Paper	p	Category	Our reading
Trigger	EW1000	89/511	17.4%	[14.4, 20.9]	0.10	2.8×10^{-7}	**Strongly inconsistent**	Clear miss
Trigger	EW2000	186/1045	17.8%	[15.6, 20.2]	0.10	1.8×10^{-14}	**Strongly inconsistent**	Clear miss
Fiducial	EW1000	33/89	37.1%	[27.8, 47.5]	0.28	0.06	**Borderline / mild tension**	Close; mild tension
Fiducial	EW2000	62/186	33.3%	[27.0, 40.4]	0.28	0.12	**Consistent**	Matches
Reconstruction	EW1000	2/33	6.1%	[1.7, 19.6]	0.72	2.0×10^{-15}	**Strongly inconsistent**	Clear miss
Reconstruction	EW2000	10/62	16.1%	[9.0, 27.2]	0.72	7.8×10^{-20}	**Strongly inconsistent**	Clear miss
Reco, no SCT z smearing*	EW1000	27/33	81.8%	[65.6, 91.4]	0.72	0.25	**Consistent**	Diagnostic only
Reco, no SCT z smearing*	EW2000	50/62	80.6%	[69.1, 88.6]	0.72	0.16	**Consistent**	Diagnostic only

* Not in the paper. Shown only to measure how much one undocumented setting matters (section 5).

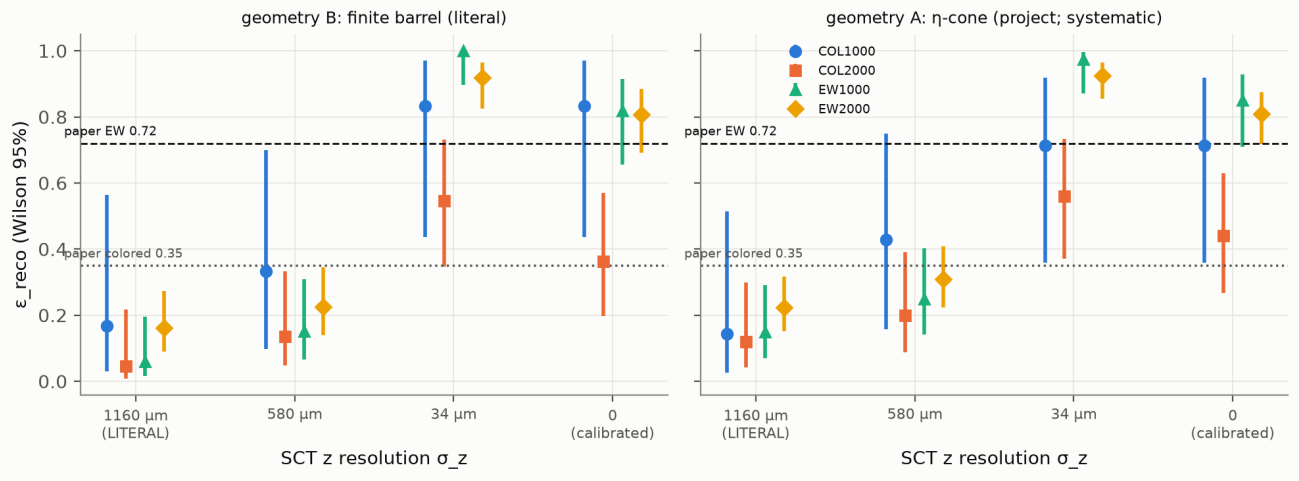


Left: EW vs colored trigger efficiency for every generator variation we tried; star = paper. Right: all EW values.

Why the EW trigger does not match

- The same trigger (quirk-pair pT > 200 GeV) gives the paper's colored value, so the definition is right.
- 36 one-at-a-time generator changes (MadGraph 2.3.3 / Pythia 8.212 from the paper's source, matching scale, PDF, scales, tune, 2 jets, 14 TeV) keep EW at 13–19%. None gets near 10%.
- The paper's EW/colored ratio is 0.42; ours stays at 0.60–0.84. Most likely an undocumented generator setting or extra cut in the paper.
- Fixing our EW model's Z coupling to vector-like (as the paper states) changes the cross-section by −24% but not the trigger.

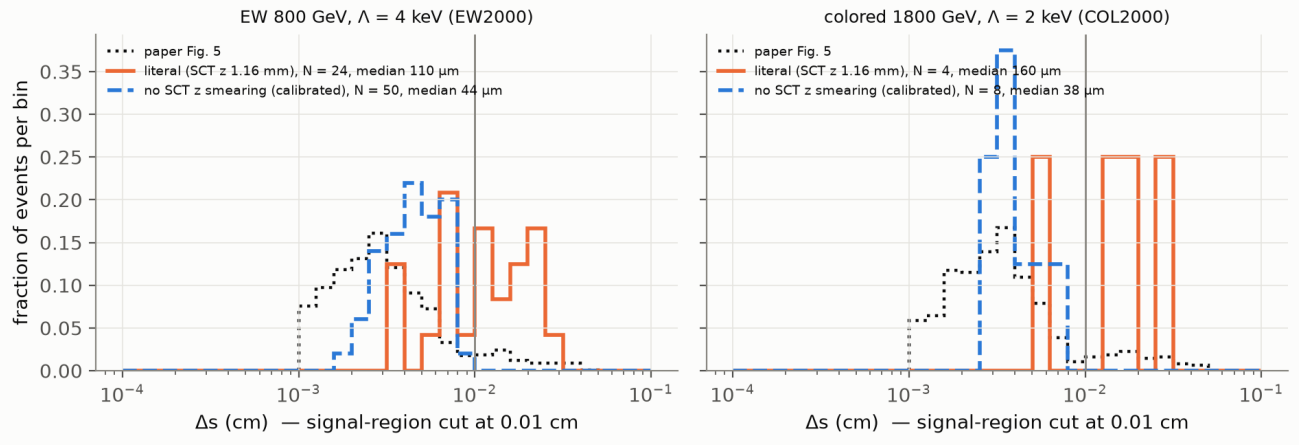
5. Why reconstruction does not match (colored and EW)



Reconstruction efficiency of the same events for different SCT resolutions along the beam. Left panel = literal geometry.

The one setting that decides it

- The paper smears SCT hits by “twice the intrinsic accuracy”. Its reference [27] gives 580 μm along the beam, so the literal value is **1.16 mm**.
- With 1.16 mm the fitted plane gets thick: our signal Δs median is 110–160 μm, above the 100 μm cut, so most events fail.
- The paper's own Fig. 5 has a signal Δs median near 30 μm (ours: 110 μm EW, 160 μm colored). That only happens with ≲ 34 μm smearing, or none. Text and figures disagree, and the paper does not say what it ran.
- Without that smearing we get EW 81–82% (paper 72%, consistent) and colored 36% for COL2000 (paper 35%). We keep this as a diagnostic because the paper does not state it.
- A fresh-seed sample, set up before generation, confirms the literal numbers: EW 14/102 = 14%, colored 6/94 = 6%.



Signal Δs before the last cut. With 1.16 mm smearing (solid) our events sit right of the paper (dotted); without it (dashed) they move close to the paper. All Fig. 5 panels incl. backgrounds: FINAL_FIGURES/F05.

6. Table I: paper vs this work

		Paper	Sample 1 (1000 gen.)	Sample 2 (2000 gen.)	Result	Main reason if not
Colored	ε_trig	0.24	0.241 [0.203, 0.284] p = 0.95	0.237 [0.210, 0.266] p = 0.87	**Reproduced**	—
	ε_fid	0.28	0.222 [0.106, 0.408] p = 0.67	0.324 [0.224, 0.442] p = 0.42	**Reproduced**	—
	ε_reco	0.35	0.167 [0.030, 0.564] p = 0.67	0.045 [0.008, 0.218] p = 0.0013	**Not reproduced**	SCT resolution: text ≠ Fig. 5
EW	ε_trig	0.10	0.174 [0.144, 0.209] $p = 2.8\times10^{-7}$	0.178 [0.156, 0.202] $p = 1.8\times10^{-14}$	**Not reproduced**	Undocumented generator settings
	ε_fid	0.28	0.371 [0.278, 0.475] p = 0.06	0.333 [0.270, 0.404] p = 0.12	**Reproduced**	—
	ε_reco	0.72	0.061 [0.017, 0.196] $p = 2.0\times10^{-15}$	0.161 [0.090, 0.272] $p = 7.8\times10^{-20}$	**Not reproduced**	SCT resolution: text ≠ Fig. 5

Values: efficiency [Wilson 95% CI], exact p vs the paper. The result column uses the larger sample; the 6-event colored sample (1/6) cannot decide anything on its own.

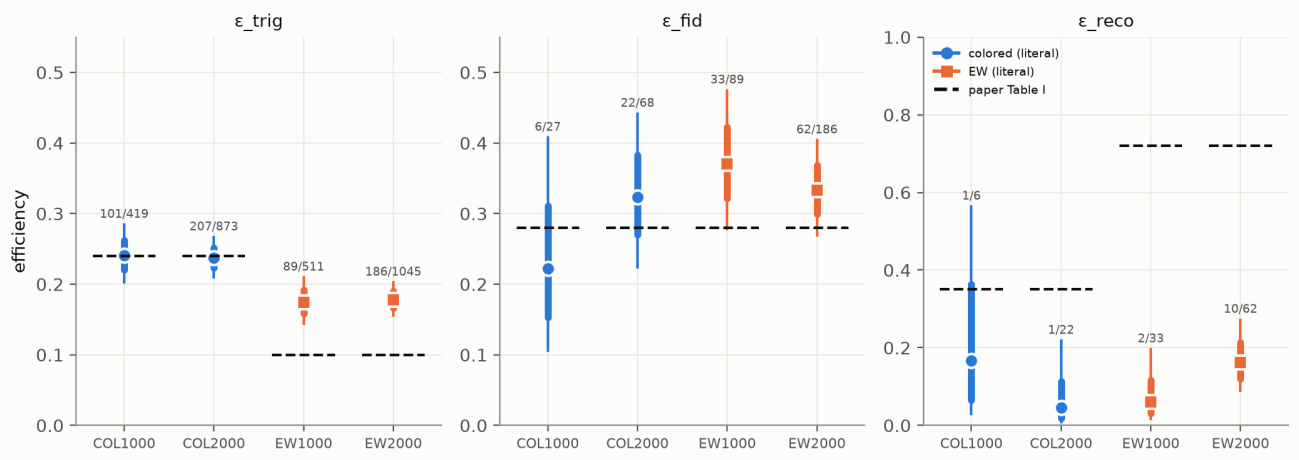


Table I comparison. Thick bar = 68%, thin bar = 95% interval; dashed line = paper.

Bottom line

- Reproduced: colored trigger, both fiducial efficiencies, and the paper's statement that no background survives the signal region.
- Not reproduced: EW trigger, and both reconstruction efficiencies with the detector exactly as the paper describes it.
- A full reproduction is not possible from the published information: the SCT resolution actually used, the generator settings, and any event counts or errors are missing.
- Matching the reconstruction numbers needs the SCT beam-axis smearing removed. That contradicts the paper's text and lets 0.24% of conversions into the signal region, so we report it only as a systematic.

Appendix A. Paper-figure reproductions

Figures 2–4 rebuilt from real simulated events (one event each, selection rule stated in each figure's source file).

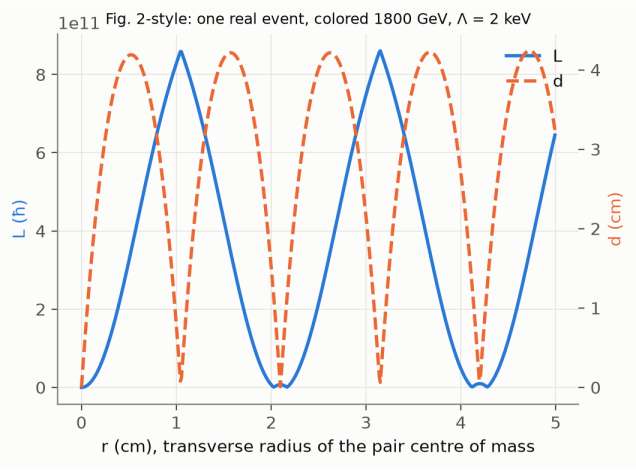


Fig. 2 style: angular momentum L and quirk separation d (colored, COL2000 event 107). L peaks near 9×10^{11} ħ, the paper's ~10^{12} ħ scale.

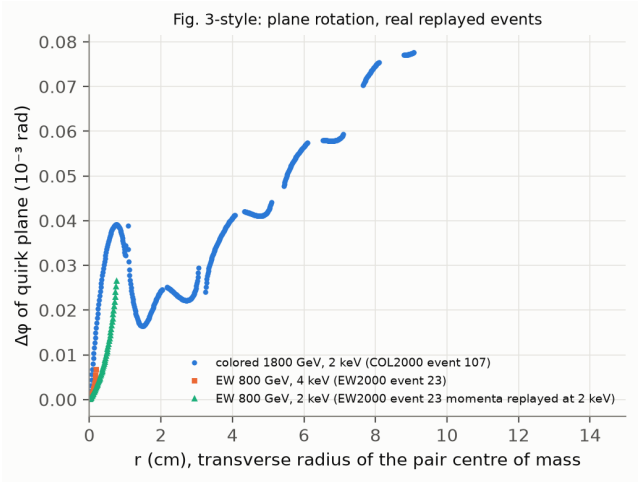


Fig. 3 style: rotation of the quirk plane stays below 10^{-4} rad, so the hits stay on a plane.

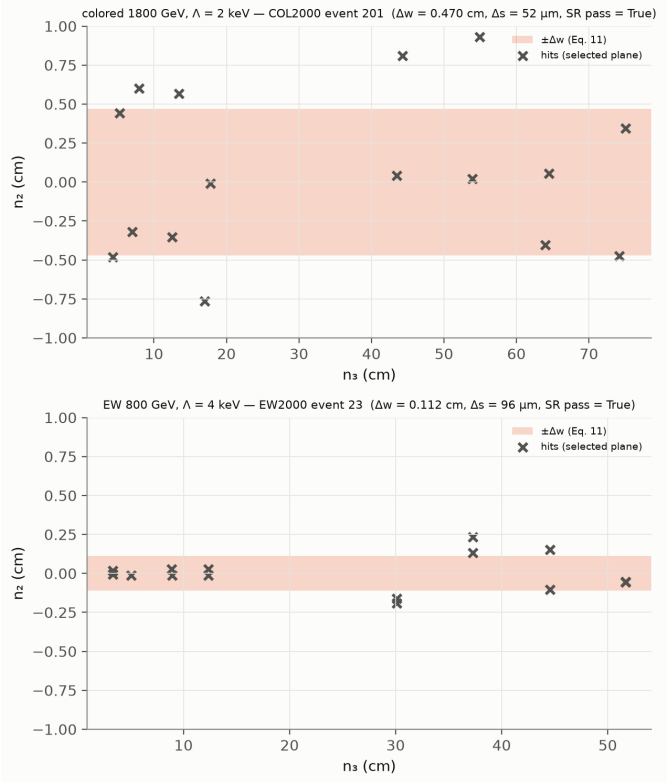


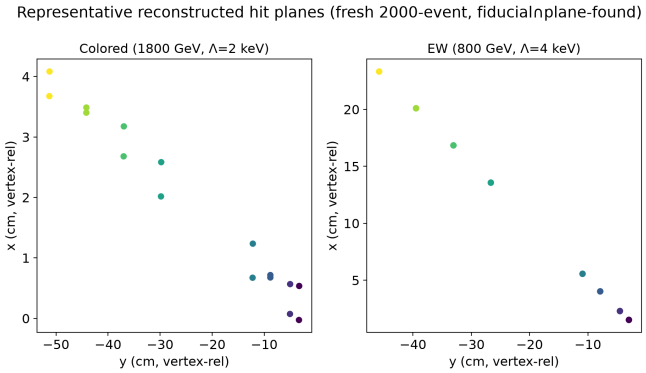
Fig. 4 style: hits of one colored and one EW event that pass the literal signal region, in the plane's own coordinates; band = ±Δw.

Appendix B. Check of the 30 September figure folder

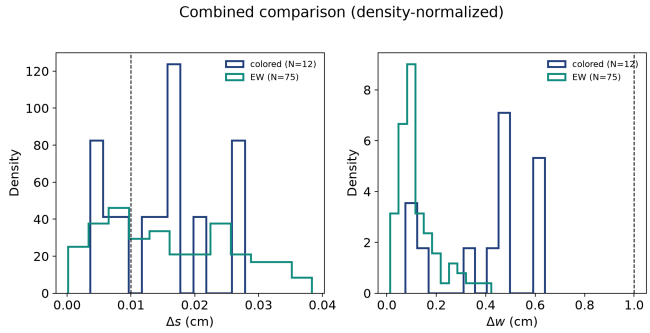
Folder `FINAL_FIGURES_2000_EVENT_2026-09-30`. Every number was recomputed from the per-event results. These figures use the older project acceptance (an $|\eta| < 1.8$ cone) instead of the paper's finite barrel, so their fiducial counts are higher than in the main report.

Item	30 Sep figure (fid / plane / mult / final)	Recomputed now	Check
COL1000	7 / 5 / 3 / 1	7 / 5 / 3 / 1	**Match**
COL2000	25 / 12 / 7 / 3	25 / 12 / 7 / 3	**Match**
EW1000	40 / 32 / 16 / 6	40 / 32 / 16 / 6	**Match**
EW2000	94 / 75 / 48 / 21	94 / 75 / 48 / 21	**Match**
Fig. 5 colored values	12 events	12/12 identical	**Match**
Fig. 5 EW values	75 events	57/75 identical; 17 differ by ≤ 30 μm and 1 has no plane on this machine (floating-point replay); 0 pass/fail changes at the Δs cut	**Match**
Fig. 4 events (65, 182)	Δs, Δw	identical	**Match**

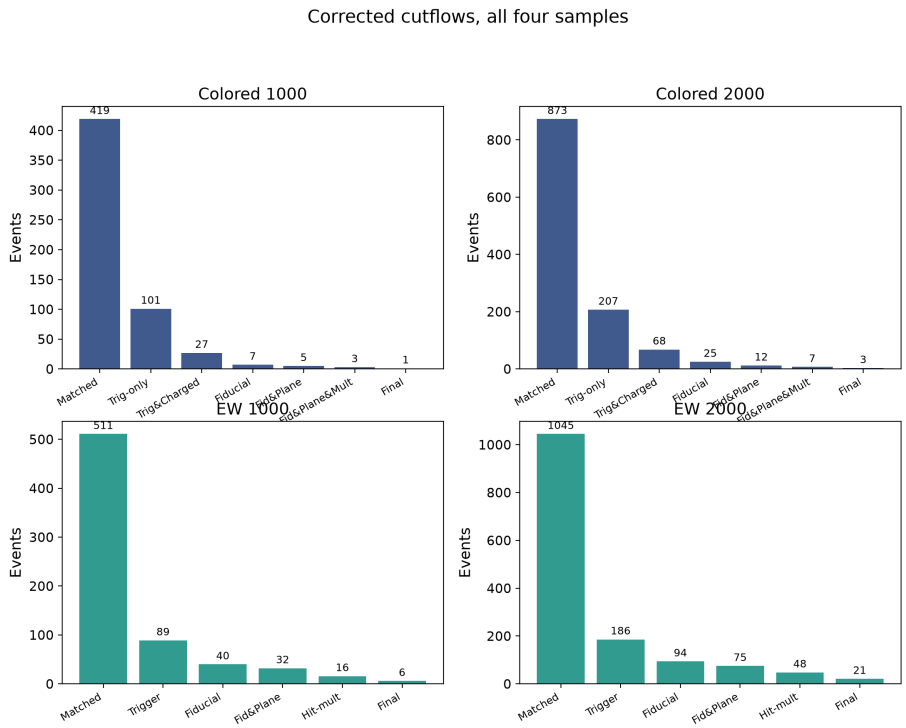
The 30 Sep Fig. 2 is not reused: it computed L with lab-frame momenta and reached ~5×10^{13} ħ. Computed in the quirk pair's rest frame (Appendix A) it is ~9×10^{11} ħ, in line with the paper.



30 Sep Fig. 4 (project acceptance): hit positions of colored event 65 and EW event 182.



30 Sep Fig. 5 (project acceptance): Δs and Δw of the 12 colored and 75 EW events with a plane.



30 Sep cutflows of all four samples (project acceptance); counts verified in the table above.